

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{2}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_d^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{36} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_d^{\mathbf{P}}] \delta_{i1i2} - \frac{8}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_d^{\mathbf{P}}] \delta_{i1i2} - \right. \\ & \frac{2}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_e^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{12} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_e^{\mathbf{P}}] \delta_{i1i2} - \frac{8}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_e^{\mathbf{P}}] \delta_{i1i2} - \\ & \frac{1}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_l^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_l^{\mathbf{P}}] \delta_{i1i2} - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_l^{\mathbf{P}}] \delta_{i1i2} - \\ & \frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_q^{\mathbf{P}}] \delta_{i1i2} - \\ & \frac{8}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_u^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_u^{\mathbf{P}}] \delta_{i1i2} - \frac{32}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_u^{\mathbf{P}}] \delta_{i1i2} - \\ & \frac{1}{9} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{6} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} + \frac{8}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i1}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i1}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i2}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i2}] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i2}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i2}] \delta_{i1i2} - \frac{1}{4} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{2,1,0} [m_1, m_l^{\mathbf{P}}] + \\ & \frac{1}{2} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{3,1,-1} [m_1, m_l^{\mathbf{P}}] - \frac{1}{4} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{4,1,-2} [m_1, m_l^{\mathbf{P}}] - \\ & \frac{3}{4} g_2^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{2,1,0} [m_2, m_l^{\mathbf{P}}] + \frac{3}{2} g_2^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{3,1,-1} [m_2, m_l^{\mathbf{P}}] - \\ & \frac{3}{4} g_2^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{4,1,-2} [m_2, m_l^{\mathbf{P}}] - \frac{1}{3} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_d^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{1}{3} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^{\mathbf{r}}, m_l^{\mathbf{P}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_e^{\mathbf{r}}, m_l^{\mathbf{P}}] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{i1}, m_1] \delta_{i1i2} - \\ & \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{i1}, m_1] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{i2}, m_1] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{i2}, m_1] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] \delta_{i1i2} + \frac{1}{3} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] \delta_{i1i2} - \\ & \frac{3}{4} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] \delta_{i1i2} + \frac{1}{3} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{2,1,0} [m_l^{\mathbf{P}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{2,2,-1} [m_l^{\mathbf{P}}, \tilde{\mu}] - \\ & \frac{1}{12} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{3,1,-1} [m_l^{\mathbf{P}}, \tilde{\mu}] + \frac{1}{3} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] \delta_{i1i2} + \\ & \frac{1}{3} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] \delta_{i1i2} - \frac{5}{4} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] \delta_{i1i2} + \\ & g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\mathbf{P}}, m_u^{\mathbf{r}}] \delta_{i1i2} - \\ & \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_u^{\mathbf{r}}] \delta_{i1i2} - \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} - \\ & \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} + \\ & g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{12} g_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} - \\ & \frac{1}{3} g_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{4} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} - g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \delta_{i1i2} - \\ & \frac{1}{2} \overline{y}_e^{\text{ps}} \overline{y}_e^{\text{ri}1} y_e^{p i 2} y_e^{\text{rs}} \text{LF}_{2,1,0} [\tilde{\mu}, m_e^{\text{s}}] + \overline{y}_e^{\text{ps}} \overline{y}_e^{\text{ri}1} y_e^{p i 2} y_e^{\text{rs}} \text{LF}_{3,1,-1} [\tilde{\mu}, m_e^{\text{s}}] - \\ & \frac{1}{2} \overline{y}_e^{\text{ps}} \overline{y}_e^{\text{ri}1} y_e^{p i 2} y_e^{\text{rs}} \text{LF}_{4,1,-2} [\tilde{\mu}, m_e^{\text{s}}] - \frac{1}{12} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{2,1,0} [\tilde{\mu}, m_l^{\mathbf{P}}] + \\ & \frac{5}{12} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_l^{\mathbf{P}}] - \frac{1}{6} g_1^2 \overline{y}_e^{p i 1} y_e^{p i 2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_l^{\mathbf{P}}] + \\ & \frac{1}{4} m_1 g_1^2 y_e^{p i 2} \overline{a}_e^{p i 1} \text{LF}_{2,1,1,0} [m_1, m_e^{i1}, m_l^{\mathbf{P}}] - 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